Foundations of OOP and Java

50 Multiple Choice Questions with Answers
Anna University – Undergraduate Students | 1 Mark Each

Topics: OOP Principles, JDK, Data Types, Variables, Arrays, Operators, Control Statements, Classes, Objects, Constructors, Access Specifiers, Static Members, Inheritance and Method Overriding.

Section 1: Principles of OOP and JDK

1. Which of the following is NOT a fundamental principle of Object-Oriented Programming?

A) Encapsulation

B) Inheritance

C) Polymorphism

D) Compilation

Answer: D) Compilation

2. Which OOP principle combines data and methods into a single unit?

A) Inheritance

B) Encapsulation

C) Abstraction

D) Polymorphism

Answer: B) Encapsulation

3. Which OOP principle allows an object to exhibit different behaviours through a common interface?

A) Abstraction

B) Encapsulation

C) Polymorphism

D) Inheritance

Answer: C) Polymorphism

4. Which component of Java is responsible for compiling source code into bytecode?

A) JVM

B) JRE

C) JDK

D) JIT

Answer: C) JDK

5. What is the primary function of the Java Virtual Machine (JVM)?

A) To write Java source code

B) To execute Java bytecode

C) To design user interfaces

D) To create Java packages

Answer: B) To execute Java bytecode

Section 2: Data Types, Variables and Arrays

6. Which of the following is a valid primitive data type in Java?

A) String

B) Array

C) boolean

D) Scanner

Answer: C) boolean

7. What is the default value of an instance variable of type int in Java?

A) 1

B) null

C) 0

D) Undefined

Answer: C) 0

8. Which of the following is the correct declaration of a character variable?

A) char ch = "A";

B) char ch = 'A';

C) character ch = 'A';

D) char ch = A;

Answer: B) char ch = 'A';

9. What is the output of int x = 10; System.out.println(x++);?

A) 10

B) 11

C) 9

D) Compilation error

Answer: A) 10

10. Which of the following is the correct way to declare an integer array in Java?

A) int arr = new int[5];

B) int[] arr = new int[5];

C) array int arr = new int[5];

D) int arr[] = int[5];

Answer: B) int[] arr = new int[5];

11. What is the index of the first element in a Java array?

A) 1

B) -1

C) 0

D) Depends on the array size

Answer: C) 0

12. What is the output of int[] a = {10, 20, 30, 40}; System.out.println(a.length);?

A) 3

B) 4

C) 5

D) Compilation error

Answer: B) 4

13. Which of the following is the default value of an instance variable of type boolean?

A) true

B) false

C) 0

D) null

Answer: B) false

Section 3: Operators and Control Statements

14. Which operator is used to compare two values for equality in Java?

A) =

B) ==

C) !=

D) ===

Answer: B) ==

15. What is the output of int a = 10, b = 3; System.out.println(a / b);?

A) 3.33

B) 3

C) 4

D) 1

Answer: B) 3

16. Which operator is known as the conditional operator in Java?

A) &&

B) ||

C) ?:

D) !

Answer: C) ?:

17. What is the output of int x = 5; System.out.println(x > 3 && x < 10);?

A) true

B) false

C) 5

D) Compilation error

Answer: A) true

18. Which control statement is used to execute a block of code repeatedly while a condition is true?

A) if

B) switch

C) while

D) break

Answer: C) while

19. Which loop executes its body at least once, regardless of the condition?

A) for

B) while

C) do-while

D) enhanced for

Answer: C) do-while

20. What is the purpose of the break statement in a loop?

A) Skips the current iteration

B) Terminates the loop

C) Restarts the loop

D) Pauses the program

Answer: B) Terminates the loop

21. Which statement skips the remaining statements in the current iteration and proceeds to the next iteration?

A) break

B) return

C) continue

D) exit

Answer: C) continue

Section 4: Classes, Objects and Constructors

22. A class in Java is best described as:

A) An instance of an object

B) A blueprint for creating objects

C) A built-in data type

D) A method that returns a value

Answer: B) A blueprint for creating objects

23. Which keyword is used to create an object in Java?

A) class

B) this

C) new

D) object

Answer: C) new

24. What is printed when an object of Student is created if Student has int age = 20 and the program prints s.age?

A) 0

B) 20

C) null

D) Compilation error

Answer: B) 20

25. Which statement about constructors is TRUE?

A) A constructor must have a return type.

B) A constructor has the same name as its class.

C) A constructor must be static.

D) A constructor cannot accept parameters.

Answer: B) A constructor has the same name as its class.

26. What is a constructor without parameters called?

A) Parameterized constructor

B) Copy constructor

C) No-argument constructor

D) Static constructor

Answer: C) No-argument constructor

27. What happens when a class does not explicitly define any constructor?

A) The program will not compile.

B) Java provides a default constructor.

C) The JVM creates an object automatically.

D) The class cannot contain instance variables.

Answer: B) Java provides a default constructor.

28. What is printed by a no-argument constructor containing System.out.println("Constructor"); when an object is created?

A) Demo

B) Constructor

C) No output

D) Compilation error

Answer: B) Constructor

29. Which keyword is used to refer to the current object in Java?

A) super

B) this

C) current

D) self

Answer: B) this

30. What is constructor overloading?

A) Defining multiple constructors with different parameter lists

B) Defining multiple classes with the same name

C) Overriding a constructor in a subclass

D) Defining a constructor with a return type

Answer: A) Defining multiple constructors with different parameter lists

Section 5: Access Specifiers and Static Members

31. Which access specifier provides the widest accessibility in Java?

A) private

B) protected

C) default

D) public

Answer: D) public

32. Which access specifier restricts access to members within the same class?

A) public

B) protected

C) private

D) default

Answer: C) private

33. What is the accessibility of a default (package-private) member?

A) Only within the same class

B) Within the same package

C) From every package

D) Only within subclasses

Answer: B) Within the same package

34. Which keyword is used to declare a class-level variable shared by all objects?

A) final

B) static

C) private

D) volatile

Answer: B) static

35. A static variable count starts at 0 and is incremented in the constructor. What is count after creating three objects?

A) 0

B) 1

C) 2

D) 3

Answer: D) 3

36. Which statement about static methods is TRUE?

A) They can directly access instance variables without an object.

B) They belong to the class rather than individual objects.

C) They must always be private.

D) They cannot accept parameters.

Answer: B) They belong to the class rather than individual objects.

37. Which keyword prevents a variable from being reassigned after initialization?

A) static

B) volatile

C) final

D) transient

Answer: C) final

Section 6: Inheritance and Method Overriding

38. Which keyword is used to implement inheritance between classes in Java?

A) implements

B) extends

C) inherits

D) super

Answer: B) extends

39. Which type of inheritance is directly supported between classes in Java?

A) Multiple inheritance

B) Hybrid inheritance

C) Single inheritance

D) Multiple inheritance through classes

Answer: C) Single inheritance

40. Which inheritance structure consists of one superclass and multiple subclasses?

A) Single inheritance

B) Multilevel inheritance

C) Hierarchical inheritance

D) Multiple inheritance

Answer: C) Hierarchical inheritance

41. Class B extends class A but does not define display(). If A defines display() to print 'Class A', what is printed by calling display() on a B object?

A) Class B

B) Class A

C) Compilation error

D) No output

Answer: B) Class A

42. Which keyword is used to access the immediate superclass constructor or members?

A) this

B) super

C) extends

D) parent

Answer: B) super

43. What is method overriding?

A) Defining methods with the same name but different parameter lists in the same class

B) Redefining an inherited method in a subclass with a compatible signature

C) Defining a method as static

D) Defining multiple constructors

Answer: B) Redefining an inherited method in a subclass with a compatible signature

44. Which annotation is commonly used to indicate that a method overrides a superclass method?

A) @Static

B) @Inherited

C) @Override

D) @Overload

Answer: C) @Override

45. Animal has sound() printing 'Animal sound'; Dog extends Animal and overrides sound() to print 'Bark'. What is printed by Animal a = new Dog(); a.sound();?

A) Animal sound

B) Bark

C) Compilation error

D) No output

Answer: B) Bark

46. Which concept is demonstrated when a superclass reference refers to a subclass object and invokes an overridden method?

A) Compile-time polymorphism

B) Runtime polymorphism

C) Encapsulation

D) Constructor overloading

Answer: B) Runtime polymorphism

47. Can a private method be overridden in a subclass?

A) Yes, always

B) Yes, using the @Override annotation

C) No, because it is not inherited by the subclass

D) Only if the subclass is abstract

Answer: C) No, because it is not inherited by the subclass

48. What happens if a subclass attempts to override a final method?

A) The method is overloaded.

B) The program compiles successfully.

C) A compilation error occurs.

D) The superclass method is deleted.

Answer: C) A compilation error occurs.

49. Class B extends A. A's constructor prints A and B's constructor prints B. What is printed when new B() is executed?

A) B A

B) A B

C) B only

D) Compilation error

Answer: B) A B

50. Which statement about constructors in inheritance is TRUE?

A) Constructors are inherited by subclasses.

B) Constructors can be overridden.

C) A subclass constructor can invoke a superclass constructor using super().

D) A superclass constructor is never executed when creating a subclass object.

Answer: C) A subclass constructor can invoke a superclass constructor using super().

Consolidated Answer Key

Q.No.	Answer	Q.No.	Answer
1	D) Compilation	26	C) No-argument constructor
2	B) Encapsulation	27	B) Java provides a default constructor.
3	C) Polymorphism	28	B) Constructor
4	C) JDK	29	B) this
5	B) To execute Java bytecode	30	A) Defining multiple constructors with different parameter
6	C) boolean	31	D) public
7	C) 0	32	C) private
8	B) char ch = 'A';	33	B) Within the same package
9	A) 10	34	B) static
10	B) int[] arr = new int[5];	35	D) 3
11	C) 0	36	B) They belong to the class rather than individual objects.
12	B) 4	37	C) final
13	B) false	38	B) extends
14	B) ==	39	C) Single inheritance
15	B) 3	40	C) Hierarchical inheritance
16	C) ?:	41	B) Class A
17	A) true	42	B) super
18	C) while	43	B) Redefining an inherited method in a subclass with a co
19	C) do-while	44	C) @Override
20	B) Terminates the loop	45	B) Bark
21	C) continue	46	B) Runtime polymorphism
22	B) A blueprint for creating objects	47	C) No, because it is not inherited by the subclass
23	C) new	48	C) A compilation error occurs.
24	B) 20	49	B) A B
25	B) A constructor has the same name as its class	50	C) A subclass constructor can invoke a superclass constr

Suggested assessment pattern: 50 questions × 1 mark = 50 marks. Each question has four options and one correct answer. Difficulty: Basic to Moderate, with theory and practical-oriented questions.